Discovery contrast

862 **Layer 1 Programme definition** genes in programme

FDR<0.05, |logFC| ≥ 0.3, endometrial expression gate; 608 up / 254 down

preregistered, hashed, and carried unchanged into Layers 3 and 4

Risk GWAS + cis-eQTL

2 / 259 **Layer 2 Colocalisation** both at 17q21.32

Gene-locus tests across 10 loci and five tissues; PP.H4 > 0.8 threshold

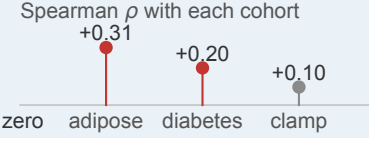
35 of 50 locus-tissue cells testable, adipose contributing one locus

(from Layer 1)

+0.31 **Layer 3 Bulk proxy validity**

Rank correlation with three independent cohorts

adipose strong, diabetes intermediate, clamp near zero



(from Layer 1)

−3.8 **Layer 4 Single-cell projection**

11 samples, 141 316 cells; tumour vs normal epithelium, sample-level test

against 100 matched random modules

Finding

One shared causal variant at 17q21.32, colocalised in liver (SKAP1, PP.H4 = 0.92) and muscle (PP.H4 = 0.87); the other nine loci are compatible with distinct causal variants

Boundary

Uterine eQTL is myometrium, not epithelium; clamp cohort 51 vs 18 limits precision